

UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2021/2022

MAY EXAMINATION

UCCN1223 CYBERSECURITY

TUESDAY, 24 MAY 2022

TIME : 9.00 AM - 11.00 AM (2 HOURS)

BACHELOR OF COMPUTER SCIENCE (HONOURS)
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
BUSINESS INFORMATION SYSTEMS
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
INFORMATION SYSTEMS ENGINEERING
BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)
COMMUNICATIONS AND NETWORKING
BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)
COMPUTER ENGINEERING

Instruction to Candidates:

This question paper consists of THREE (3) questions in Section A and TWO (2) questions in Section B.

Answer **ALL** questions in **Section A** and **ONLY ONE (1)** question in **Section B**.
Each question carries 25 marks.

Should a candidate answer more than ONE (1) question in Section B, marks will only be awarded for the FIRST question in that section in the order the candidate submits the answers.

Candidates are allowed to use a calculator.

Answer questions only in the answer booklet provided.

UCCN1223 CYBERSECURITYSECTION A (Compulsory Questions)

- Q1. (a) Alisaie sends a note to Alphinaud which prints “**Znk Cgxxoux ul Romnz tkkj nkrvy. Yktj g inuiuhu zu nkx.**”. She is using Caesar Cipher to encrypt this message. Decrypt the note. (6 marks)
- (b) Asymmetric cryptography is a cryptographic system that uses pairs of keys. Each pair consists of a public key and a private key. The public key will be shared to the other users who would like to encrypt and decrypt data for data exchange purposes. While the private key should be kept secretly, and it must not be possible for other users to access someone’s private key. Answer the following questions:
- (i) Explain the differences between file encryption and digital signature. (6 marks)
- (ii) Name **ONE (1)** algorithm that uses asymmetric cryptography. (2 marks)
- (c) Given the following 2x2 Hill cipher encryption key. Assumed that ‘A’ = 0. Encrypt the plaintext ‘JOHN’. (5 marks)
- $$\begin{pmatrix} 23 & 10 \\ 5 & 21 \end{pmatrix}$$
- (d) Determine the type of attack mentioned based on the description given below:
- (i) Exhausting the resources and bandwidth of a system, server, or network with traffic. (1 mark)
- (ii) Information gathering by inserting an anonymous into a two-party communication. (1 mark)
- (iii) An anonymous trying to send some gift cards to a database administrator and expects him/her to retrieve the list of customer information from the organization. (1 mark)
- (iv) An email from alpha_bank@yahoo.com reminding a user to change his username and password in the attached link. (1 mark)
- (v) A program that is installed unintentionally and blocks user’s access to the files and folders. (1 mark)
- (vi) A pharming attack directed at changing a host’s Domain Name Server (DNS) cache. (1 mark)

[Total : 25 marks]

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Q2. (a) With the leap toward a digital world, the banks implemented some methods to keep the users from financial fraud and scams. Discuss **TWO (2)** methods used by the banking system for defending against credit card fraud. (6 marks)

(b) Avacado has just signed up for a Cisco Networking Academy account online. He has secured the account with the following password:

#U1c2C1n3

(i) Compute the complexity of Avacado's password. (4 marks)

(ii) The Cisco Networking Academy account strengthens the password for each user with an 8-bit salt. Calculate the complexity of Avacado's salted password. (3 marks)

(c) Scenario: Venus has the root access to her Unix system, and she created a file name as 'HelloEorzea.py'. She has the full access including execute the file. Other than herself, her group member only can view and write the contents of the file. Everyone else is unable to open and view the contents of the file.

(i) Show the permission for the file 'HelloEorzea.py' in the format of both 10-characters notation, including the Unix command to show the permission. (4 marks)

(ii) Suppose Venus wants to change the permission for the file named "HelloEorzea.py" to only allow herself to have full access, but everyone else would not be able to perform anything to the file. Write the Unix command to set the appropriate permission on the file and show the new permission for the file "HelloEorzea.py" in the format of both 10-character and octal notation. (8 marks)

[Total : 25 marks]

Q3. (a) John would like to send a confidential information to his client by embedding a secret message within an image file by using digital watermarking. However, Derrick suggests his to use steganography instead. Examine which process is more suitable in this scenario. Explain why. (5 marks)

(b) Intrusion Detection System (IDS) is designed to detect unauthorized activity within inner network, or on individual machines. Describe **THREE (3)** types of network attacks that could be detected by the IDS. (9 marks)

UCCN1223 CYBERSECURITY**Q3. (Continued)**

- (c) Identify the type of alarms (e.g., true negative) for the following scenarios.
- (i) A fire alarm is not activated when there is a fire. (1 mark)
 - (ii) The IDS do not sound an alert when there are no suspicious activities in the network. (1 mark)
 - (iii) The firewall block malicious software. (1 mark)
- (d) Antivirus is a kind of software used to prevent, scan, detect and delete viruses from a computer. There are two kinds of antivirus, which are Conventional Antivirus and Cloud-Based Antivirus. Discuss what are advantages and disadvantages of Cloud-Based Antivirus over Conventional Antivirus.

(8 marks)

[Total : 25 marks]

SECTION B (Answer Any One Question)

- Q4. (a) Figure 4 shows a firewall that is configured in Router0 in which only TCP established is allowed between network 179.60.194.35/24 and 42.189.13.42/24. All other traffic is blocked by the firewall. Based on the information given, answer the following:

- (i) Name the type of firewall setup given. (2 marks)
- (ii) Write an ACL for the firewall configuration mentioned in Q4.(a)(i). (5 marks)

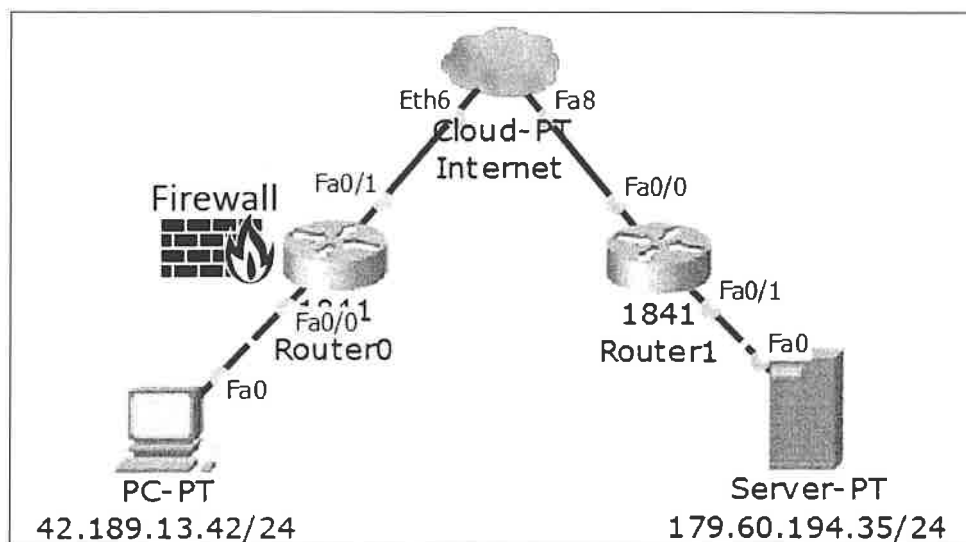


Figure 4: Firewall Configuration

UCCN1223 CYBERSECURITY**Q4. (Continued)**

- (b) Case study: Lisa has accidentally deleted a folder which consists of some important documents in her company's computer. She tried to recover the folder but only some files could be recovered partially.
- (i) Discuss why it was possible to recover some of the deleted files. (4 marks)
 - (ii) Suggest **ONE (1)** possible reason why some of the files could only be recovered partially. (2 marks)
- (c) When an attack occurs in a server and the server needs to be secured, it may be incorrect to power off the server for investigation purposes. Discuss **FOUR (4)** reason why it is not recommended to power off the server. (8 marks)
- (d) Wired Equivalent Privacy (WEP) became widely insecure. List **FOUR (4)** steps to secure an organization wireless network. (4 marks)
- [Total : 25 marks]

- Q5. (a) Figure 5a shows a network topology with a Site-to-Site IPsec VPN configured in between R2 and R4. Figure 5b is the return result in R4 after "**show crypto ipsec sa**" command is issued. Based on the information given, answer the following:
- (i) The crypto map is configured in which interface of R4? (2 marks)
 - (ii) What is the peer's IP address of R4? (2 marks)
 - (iii) Which service will be required to use VPN? (2 marks)
 - (iv) There are five (5) packets is delivered from R4 to R2, but only 2 packets are encrypted. Explain why. (2 marks)
 - (v) Suggest **ONE (1)** way to overcome the situation stated in Q5(a)(iv). (2 marks)

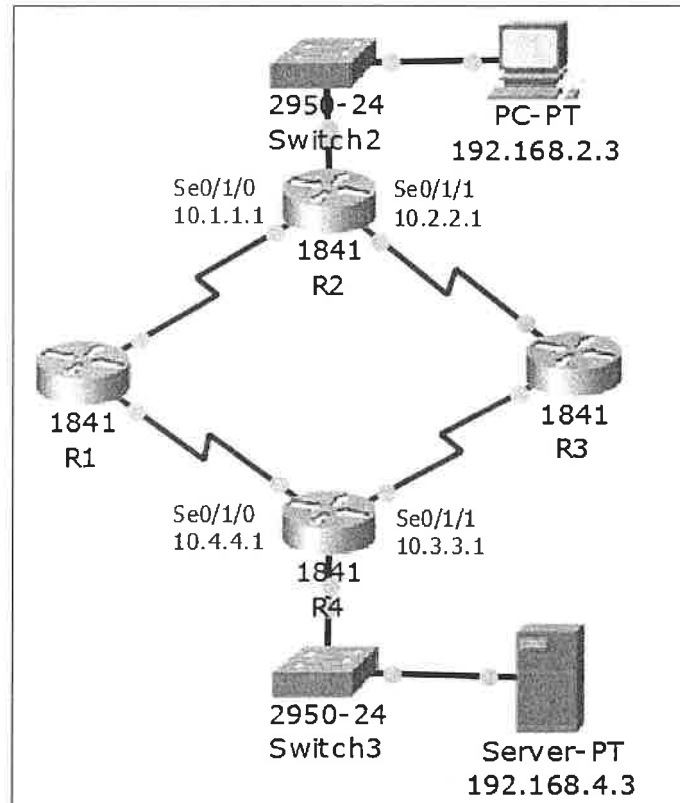
UCCN1223 CYBERSECURITYQ5. (a) (Continued)

Figure 5a: A Site-to-Site IPsec VPN configured network topology

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R4#show crypto ipsec sa

interface: Serial0/1/0
  Crypto map tag: VPN-MAP, local addr 10.4.4.1

  protected vrf: (none)
  local ident (addr/mask/prot/port):
(192.168.4.3/255.255.255.255/6/21)
  remote ident (addr/mask/prot/port):
(192.168.2.0/255.255.255.0/6/0)
  current_peer 10.1.1.1 port 500
    PERMIT, flags={origin_is_acl,}
    #pkts encaps: 2, #pkts encrypt: 2, #pkts digest: 0
    #pkts decaps: 3, #pkts decrypt: 3, #pkts verify: 0
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0

    local crypto endpt.: 10.4.4.1, remote crypto endpt.:10.1.1.1
    path mtu 1500, ip mtu 1500, ip mtu idb Serial0/1/0
    current outbound spi: 0x66FA5947(1727682887)

  inbound esp sas:
    spi: 0x28C133C8(683750344)

```

Figure 5b: The return result in R4.

UCCN1223 CYBERSECURITY**Q5. (Continued)**

- (b) A password policy is a set of rules designed to enhance company security. Recommended **FOUR (4)** best practices for password policy. (4 marks)
- (c) After the completion of ranked vulnerability risk worksheet, a risk control must be performed. Identify **FIVE (5)** possible strategies to control risks. (5 marks)
- (d) Packet sniffers “read” information traversing a network passively by only collecting data, but not attempt “entry” to “steal” data. This causes them extremely hard to be detected. Answer the following questions:
- (i) Suggest **TWO (2)** methods that can be used to detect sniffers in a network. (4 marks)
- (ii) Suggest a way to overcome packet sniffing by providing an example. (2 marks)
- [Total : 25 marks]
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